

Molecule Data Structure

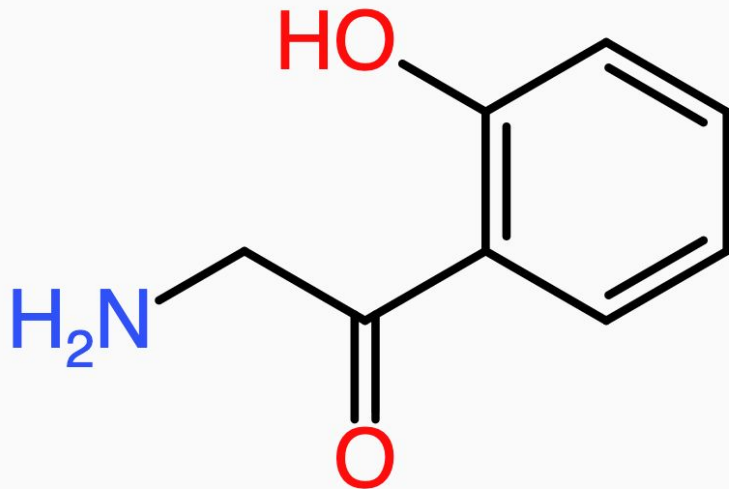
Advanced DSA - CS 03

By Ragavishagan Rajasekar
20 June 2026

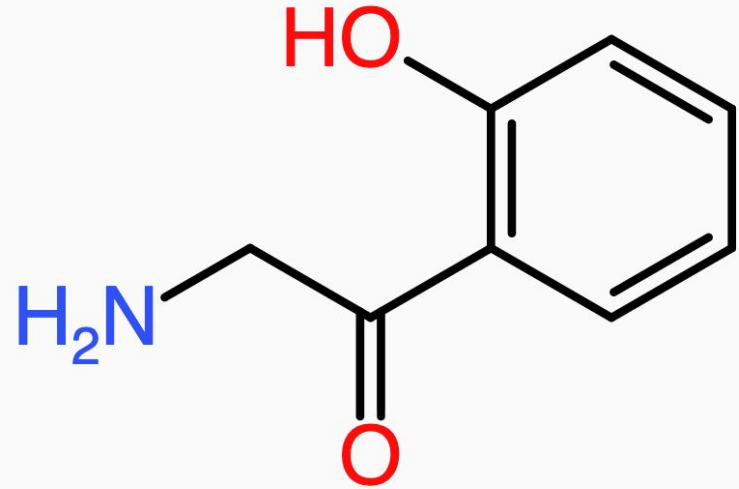
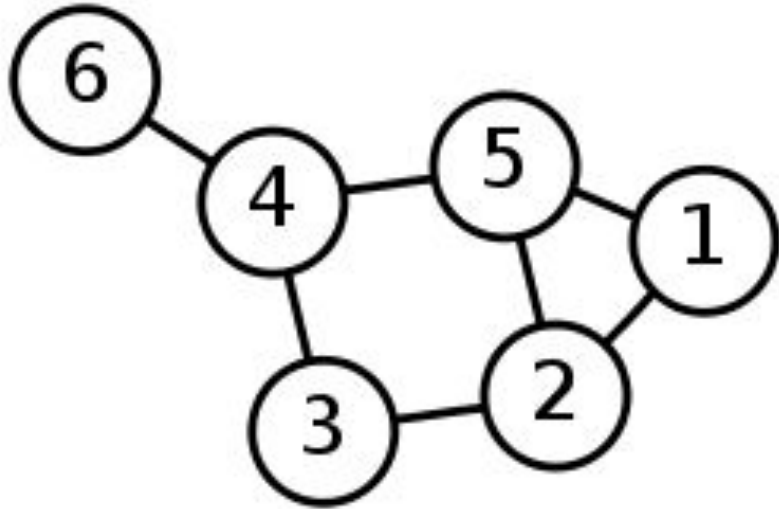
What kind of data structure can we use to represent molecules?

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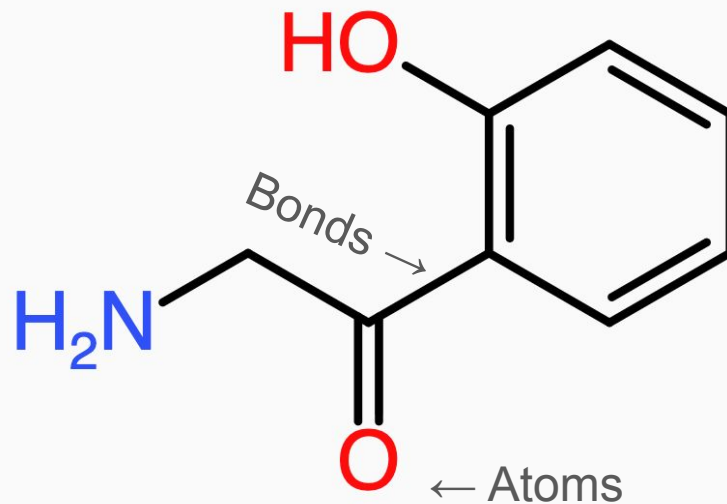
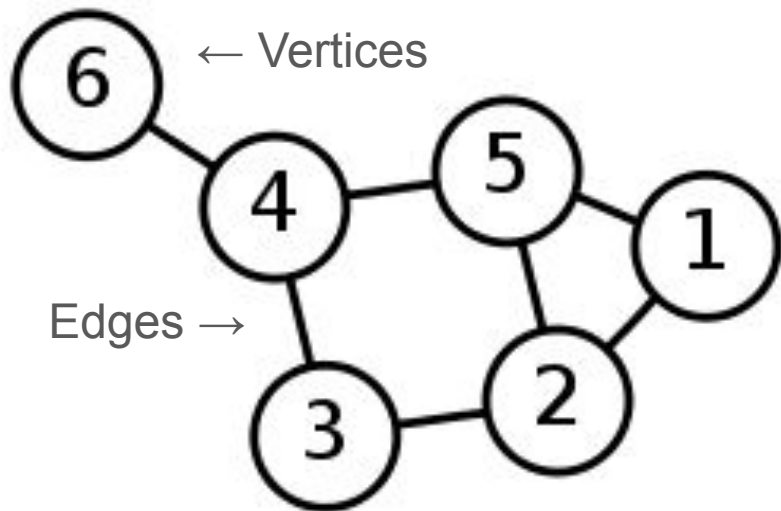
What does this remind me of??



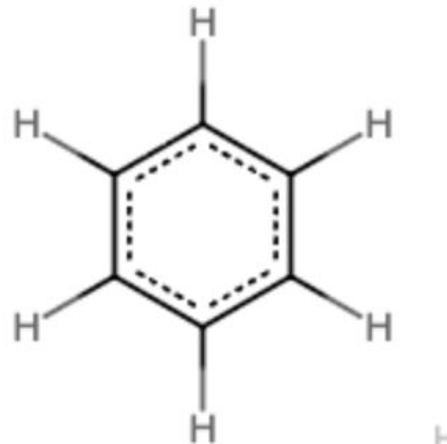
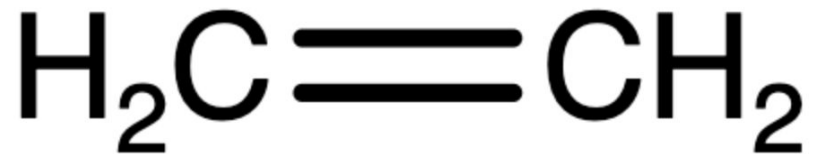
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Different types of Bonds



What is a SMILES?

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Simplified Molecular Input Line Entry System

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“CCO”



“CCC”

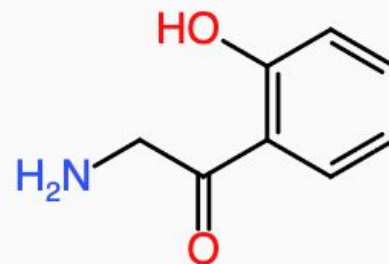
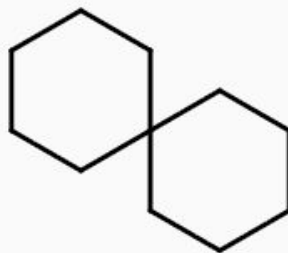
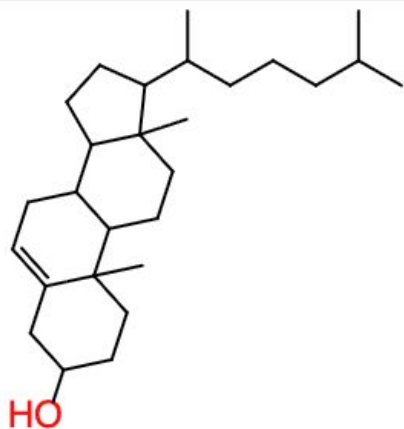
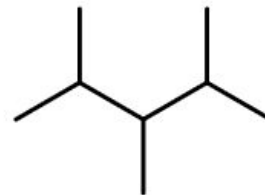
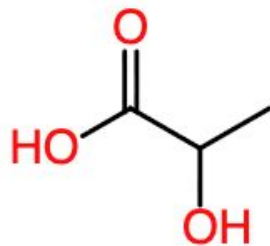


More Smiles Examples

```

1 CC(C(=O)O)O
2 C12(CCCCC2)CCCCC1
3 CC(C(C(C)C)C)C
4 c1cccc(O)c1C(=O)CN

```



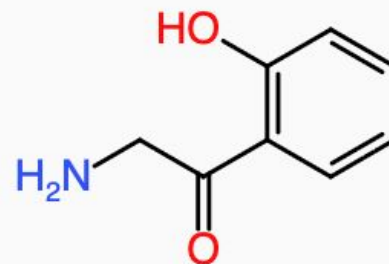
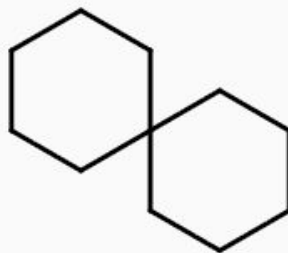
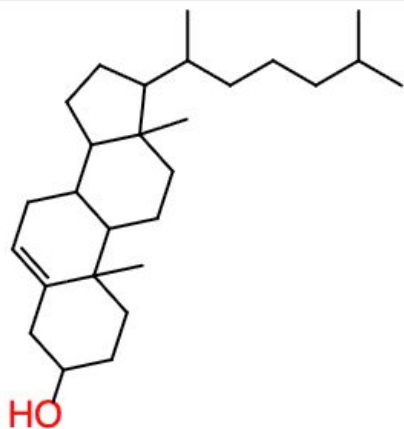
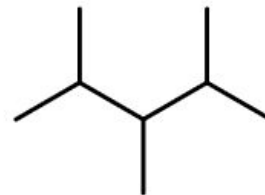
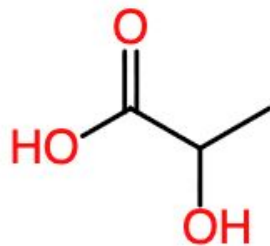
“CC(C)CCCC(C)C1CCC2C1(CCC3C2CC=C4C3(CCC(C4)O)C)C”

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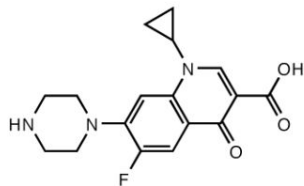
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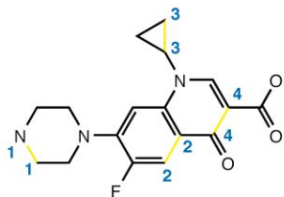
“CC(C)CCCC(C)C1CCC2C1(CCC3C2CC=C4C3(CCC(C4)O)C)C”

How do SMILES relate to graphs?

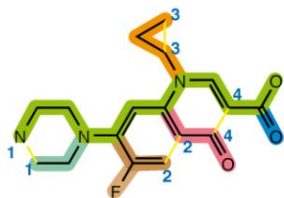
A



B



C



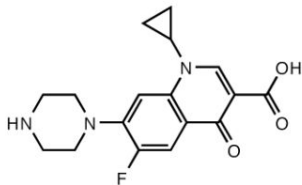
D

N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O



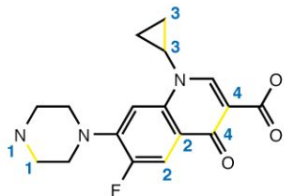
How do SMILES relate to graphs?

A

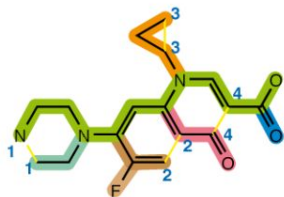


← This is a graph

B



C

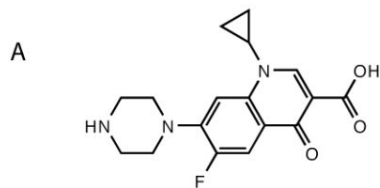


D

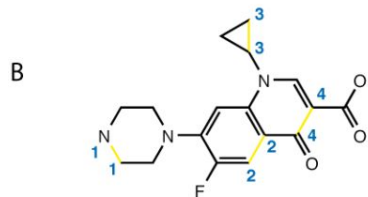
N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O



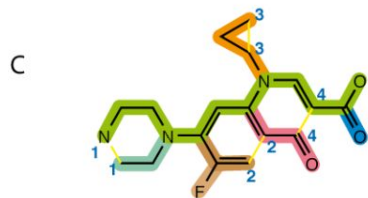
How do SMILES relate to graphs?



← This is a graph



← This is a MST (Kruskal)



← This is a Depth First Search

D

N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O



SMILES terminology

Simple Terminology

“C-C” or “CC” = Single bond

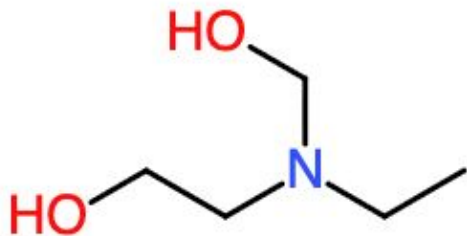
“C:C” or “cc” = Aromatic bond

“C=C” = Double bond

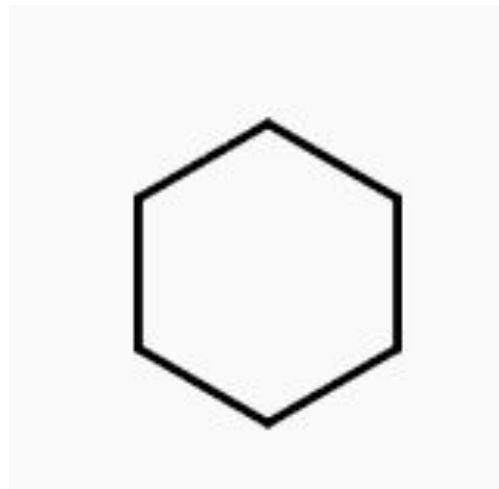
“C#C” = Triple Bond

Parenthesis indicate a Branch

CCN(CCO)CO



Numbers indicate a ring
C1CCCCC1

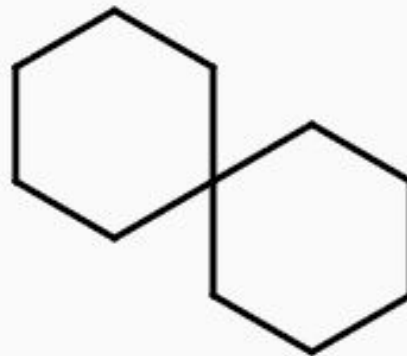


More Complex SMILES Terminology

C1CCCCC=1



C12(CCCCC2)CCCCC1



Ok I can represent a molecule as a graph, but what practical purpose does it serve?

Purpose of Molecule Graph

- Storing Data
- Working with molecules in code (Rdkit, Bioinformatics)
- Using Graph algorithms
 - Create SMILES
 - MST and DFS
 - Predicting Boiling Point
 - Floyd–Warshall algorithm applied to the Wiener Index

Wiener Index and Floyd–Warshall algorithm

Let's look at their definitions ...

Wiener Index : “In chemical graph theory, the Wiener index introduced by Harry Wiener, is a topological index of a molecule, defined as the sum of the lengths of the shortest paths between all pairs of vertices in the chemical graph representing the non-hydrogen atoms in the molecule”

Floyd–Warshall algorithm: “is an algorithm for finding shortest paths ... A single execution of the algorithm will find the lengths (summed weights) of shortest paths between all pairs of vertices.”

Almost exactly the same thing.

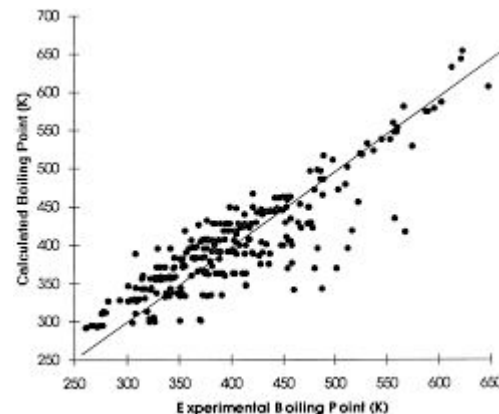
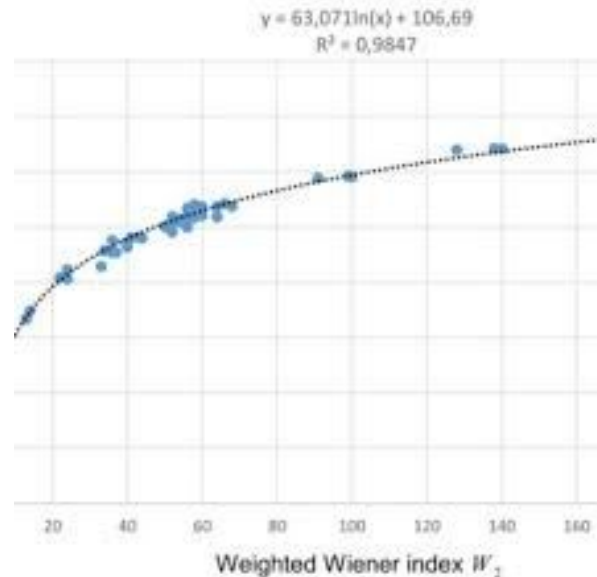
Application of Wiener's Index

The Wiener index of a molecule can be used to predict the boiling point of molecules.

$$BP = \alpha \ln(W) + \beta$$

Is a general formula where alpha and beta are constants depending on the family of molecules and W is the Wiener Index.

The formula is not too accurate when other intermolecular forces are present but the formula can be tweaked with correction factors to more accurately determine the boiling point.



Conclusion

Thank you for watching my presentation to the very end!

There's definitely a lot more I can do with this.

Check out RDKit to see what the current standard Cheminformatics library can accomplish with graph representations of molecules.